
Within-Group Variability in Automated Speech Recognition of African American English: Evidence from CORAAL DCA

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Abstract

We evaluate within-group variability in automated speech recognition (ASR) performance for African American English (AAE) speakers using wav2vec2-large-960h on the CORAAL DCA subcorpus ($N = 68$ speakers, 58,552 utterance-level observations). The presence of a 4-gram KenLM language model produces the largest reduction in Word Error Rate (WER), from $M = 0.401$ to $M = 0.309$ (Cohen’s $d = 2.22$, $p \ll 0.001$). Subcommunity membership predicts significant WER variation ($\eta^2 = 0.042$), with Ward 3 speakers showing the lowest error rates. Age group ($\eta^2 = 0.024$) and phonetic reduction ($d = 0.07$) contribute smaller but significant effects. Gender is not significant in mixed-effects models. We situate these findings within a methodological framework for disentangling ethnic linguistic variation from regional dialect effects, addressing the “confounding by provenance” problem. The study establishes baseline within-group variability benchmarks necessary for interpreting cross-racial ASR disparity estimates.

Keywords automated speech recognition; African American English; Word Error Rate; sociophonetics; algorithmic fairness; CORAAL

1 Introduction

A speaker asks a voice assistant for directions; the system transcribes a different city. A job candidate dictates a cover letter into her phone; the output contains word substitutions that change the meaning of key sentences. These failures are not distributed evenly across the population. Koenecke et al. (2020) demonstrated that five major commercial ASR systems—from Apple, Amazon, Google, IBM, and Microsoft—produced approximately twice the Word Error Rate (WER) for African American speakers compared to White speakers, a finding that has since been replicated across neural architectures including self-supervised models such as wav2vec 2.0, HuBERT, and Whisper (Chang et al., 2024; Jahan et al., 2025). The downstream consequences extend beyond inconvenience: Mengesha et al. (2021) documented that African American users experience identity threat and engage in speech accommodation when ASR systems fail on their speech, and Cunningham et al. (2025) articulated a governance framework identifying both allocative and representational harms.

The magnitude and the interpretation of these disparities, however, rest on an unresolved methodological problem. Li et al. (2024) showed that the primary corpus used to document ASR racial bias—the Corpus of Regional African American Language (CORAAL; Kendall and Farrington 2018)—contains systematic between-site differences in recording quality, a problem they term “confounding by provenance.” When recording quality and geographic region are not held constant, apparent racial disparities in WER conflate linguistic variation with technical audio differences. Only one study to date, Scott et al. (2025), has compared ASR performance across ethnic groups from the same geographic region—the Pacific Northwest—using the PNWE corpus, and their analysis was limited to a single region. No large-scale study has systematically evaluated ASR performance for African American and White speakers drawn from the same region with

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matched recording conditions, nor has any study established baseline estimates of within-group variability among AAE speakers against which cross-racial comparisons can be interpreted.

We address one dimension of this gap by examining within-group variability in ASR performance among 68 African American English speakers from the CORAAL Washington DC subcorpus (CORAAL DCA), evaluated on wav2vec2-large-960h (Mojarad, 2025). Using 58,552 utterance-level WER observations across two ASR conditions—with and without a 4-gram KenLM language model—we identify which linguistic and demographic factors predict ASR error rates within a single ethnic group, a single region, and a single elicitation context. Our central finding is that the ASR condition—specifically, the presence or absence of the language modeling component—dominates all demographic and linguistic predictors in magnitude, reducing mean per-speaker WER by 23% (Cohen’s $d = 2.22$). Among speaker-level factors, subcommunity membership ($\eta^2 = 0.042$) and age group ($\eta^2 = 0.024$) emerge as significant predictors, while phonetic reduction, as classified by Montreal Forced Aligner alignment, contributes a small but detectable effect ($d = 0.07$). Speaker gender is not significant in the mixed-effects model.

These findings advance the literature in three ways. First, we provide the first quantitatively rigorous decomposition of within-group WER variability among AAE speakers, establishing variance components that any future cross-racial study must account for. The intraclass correlation coefficient of 0.196 indicates that approximately 20% of utterance-level WER variance lies between speakers—a substantial clustering effect that motivates hierarchical modeling. Second, we demonstrate that the language model component of a modern end-to-end ASR system exerts a considerably larger influence on error rates than any single speaker-level demographic variable, a finding with practical implications for ASR fairness interventions. Third, we articulate a methodological framework—region-fixed sampling, recording quality covariate modeling, and nested demographic comparisons—that addresses the confounding-by-provenance critique and can guide future multi-ethnic ASR evaluation studies.

2 Literature Review

2.1 Racial Disparities in ASR: The Core Evidence

Koenecke et al. (2020) established the empirical benchmark for ASR racial disparity research, reporting that five commercial systems produced WER approximately twice as high for African American speakers as for White speakers, with error rates positively correlated with the density of AAE phonological and morphosyntactic features in the speech sample. The study analyzed structured interviews from CORAAL and a matched sample from the Corpus of Contemporary American English, covering multiple US cities. Its impact on the algorithmic fairness and speech technology communities has been substantial, but its geographic sampling was not regionally balanced: African American speakers were drawn predominantly from Washington DC, Princeville (NC), and Rochester (NY), while White speakers came from different regions with different recording equipment and collection protocols. Chang et al. (2024) replicated the core disparity finding across self-supervised learning models—wav2vec 2.0, HuBERT, WavLM, and XLS-R—demonstrating that the AAVE-MAE performance gap persists even in architectures trained on hundreds of thousands of hours of unlabeled speech. Using the Fair-Speech benchmark dataset of 26,500 prompted utterances from 593 speakers (Velic et al., 2024), they further showed that AAVE morphosyntactic and phonological feature density independently predicts WER, confirming that the error patterns are linguistically patterned rather than stochastic. Jahan et al. (2025) extended the evidence to podcast and interview speech via the FaiST benchmark, finding bias across 18 ASR systems, though their White comparison sample was drawn from a separate corpus (VoxCeleb), introducing a potential domain confound.

2.2 Linguistic Mechanisms and the Regional-Ethnic Confound

The mechanisms driving these disparities have received increasing attention. Scott et al. (2025) provided the most fine-grained sociophonetic analysis to date, linking specific phonological variables—including the low-back merger and pre-nasal vowel patterns—to ASR error rates across four ethnic groups (African American, Caucasian American, ChicanX, and Yakama) from the Pacific Northwest English (PNWE) corpus. Their use of the same geographic region for all ethnic groups represents a critical methodological advance, demonstrating that ethnic linguistic variation can produce ASR disparities even when regional dialect is held constant. However, the PNWE corpus is modest in size (~ 80 speakers total), limiting statistical power for interaction effects, and the single-region design leaves open the question of whether the Pacific Northwest findings generalize to other US regions with different patterns of AAE feature distribution. Seditova and Tang (2026) analyzed over 3,000 ASR errors from the DECTE Newcastle corpus and classified them by

linguistic domain, finding that phonology accounted for the largest share—consistent with Scott et al.’s emphasis on phonetic and phonological variables—though morphosyntactic and lexical factors also contributed. O’Neill and Carson-Berndsen (2023) similarly documented systematic phoneme-level substitution patterns in ASR errors on non-standard speech.

2.3 The Confounding-by-Provenance Problem

The study most directly motivating the present work is Li et al. (2024), who re-examined the Koenecke et al. (2020) findings with a focus on recording quality. They demonstrated that between-site differences in CORAAL’s recording equipment, environment, and digitization parameters create a spurious “confounding by provenance” effect: when recording quality is not modeled, apparent racial disparities in WER conflate genuine linguistic variation with technical audio differences across collection sites. This finding does not invalidate the existence of ASR racial disparities—the disparities persist even after partial control—but it establishes that prior estimates of the disparity magnitude may be inflated and that experimental designs must account for recording quality as a covariate. The implication is clear: the strongest evidence for racial disparities in ASR would come from a study in which African American and White speakers are recorded with identical equipment in the same physical setting within the same geographic region, or in which recording quality metrics are explicitly modeled as covariates.

2.4 What Remains Unresolved

Despite this body of work, three questions remain open. First, no study has established how much WER variation exists within African American English speakers alone, after controlling for region—without baseline within-group variance estimates, interpreting cross-group differences is inherently speculative. Second, the relative importance of the ASR system’s internal components (acoustic model, language model, decoder) versus speaker-level demographic and linguistic factors has not been quantitatively decomposed; most prior work treats “ASR system” as a black box. Third, the contribution of utterance-level phonetic reduction—as distinct from stable phonological dialect features—to ASR errors within AAE speech has not been isolated. Reduction processes such as consonant cluster simplification and vowel centralization are characteristic of spontaneous speech generally and AAE specifically (Green, 2002; Wolfram and Schilling, 2016), but their independent effect on ASR error rates, separate from demographic variability, remains unmeasured. The present study addresses these three gaps through a within-group analysis of the CORAAL DCA subcorpus evaluated on wav2vec2-large-960h.

3 Data

3.1 Corpus and ASR Evaluation Data

We use the CORAAL DCA (Washington DC) subcorpus of the Corpus of Regional African American Language (Kendall and Farrington, 2018), accessed through an ASR evaluation dataset prepared by Mojarad (2025) and deposited on the Open Science Framework (OSF project qn6a2). The underlying corpus consists of spontaneous sociolinguistic interviews conducted between 1968 and 1969 with African American residents of Washington DC. The ASR evaluation dataset provides utterance-level Word Error Rate (WER) computed from wav2vec2-large-960h (Baevski et al., 2020) in two conditions: with a 4-gram KenLM language model trained on LibriSpeech and without any language model, using greedy decoding in both cases. Each observation represents a single target word from the CORAAL orthographic transcription, evaluated against the ASR hypothesis for the corresponding utterance; WER is computed per utterance via the standard formula (Substitutions + Deletions + Insertions) divided by the number of reference words.

3.2 Sample

The dataset contains 58,552 utterance-level observations from 68 African American English speakers (40 male, 28 female) distributed across four age groups—ag1: under 20 ($N = 38$ speakers), ag2: 20–29 ($N = 13$), ag3: 30–49 ($N = 11$), ag4: 50+ ($N = 6$)—and three subcommunities within Washington DC: Ward 1 / NW DC (se1, $N = 20$ speakers), Ward 2 / NE-SE DC (se2, $N = 21$), and Ward 3 / SE DC (se3, $N = 27$). Each of the 29,276 unique utterances appears under both ASR conditions (with and without language model), yielding a balanced within-subject design for the ASR condition factor. Speaker-level demographic metadata—year of birth, age, education level, occupation, and neighborhood quadrant—were enriched by

Table 1: Descriptive statistics for key variables in the CORAAL DCA ASR evaluation dataset.

Variable	<i>N</i> (utterances)	Mean WER	Median WER	<i>SD</i>
Overall	58,552	0.340	0.300	0.273
ASR condition				
quad With LM	29,276	0.296	0.250	0.263
quad Without LM	29,276	0.383	0.348	0.276
Gender				
quad Male	38,142	0.341	0.300	0.270
quad Female	20,410	0.337	0.300	0.279
Age group				
quad ag1 (<20)	31,862	0.367	0.333	0.283
quad ag2 (20–29)	11,292	0.280	0.235	0.250
quad ag3 (30–49)	10,504	0.292	0.250	0.242
quad ag4 (50+)	4,894	0.399	0.368	0.278
Subcommunity				
quad se1 (Ward 1)	13,606	0.404	0.375	0.292
quad se2 (Ward 2)	20,882	0.373	0.333	0.268
quad se3 (Ward 3)	24,064	0.274	0.222	0.252
MFA status				
quad Original	42,530	0.334	0.294	0.274
quad Reduced	16,022	0.353	0.333	0.269

merging with the CORAAL SPADE sibilants analysis dataset (OSF project 7rj6q), which provides metadata for 51 of the 68 speakers.

3.3 Variables

The primary dependent variable is utterance-level Word Error Rate (WER), a continuous measure bounded between 0 and 1, with an overall mean of 0.340 (median = 0.300, *SD* = 0.273). The distribution is positively skewed (skewness = 0.653) with moderate right-tail mass at WER = 1.0, corresponding to utterances for which wav2vec2 produced empty output (4.6% of observations). The independent variables are: ASR condition (`with_lm`, `without_lm`; within-subject), speaker gender (Male, Female; between-subject), age group (ag1–ag4; between-subject), subcommunity (se1–se3; between-subject), and MFA pronunciation status (Original Pronunciation, Reduced Pronunciation; within-subject, as classified by Montreal Forced Aligner phone-level alignment). We also record interviewer ethnicity (African American, White) as a potential speech accommodation covariate, though interviewer metadata fields were empty for all 68 speakers in this dataset—a known data limitation addressed in the Discussion.

4 Methodology

4.1 Research Design

We adopt a within-group corpus evaluation design with two analytical levels: utterance-level hypothesis tests that treat each observation as an independent data point, and a hierarchical linear mixed-effects model that accounts for the nested structure of utterances within speakers. The design is observational—speaker demographics are pre-existing, not experimentally assigned—and the ASR condition factor is experimentally controlled as a within-subject manipulation. The central analytical objective is to decompose the variance in WER attributable to the ASR system’s internal components (language model vs. no language model) versus speaker-level demographic and linguistic factors, using data from a single ethnic group (African American), a single region (Washington DC), and a single elicitation context (spontaneous sociolinguistic interview) to eliminate the regional and recording-quality confounds that complicate cross-racial comparisons.

4.2 Statistical Approach

We conduct seven hypothesis tests at the utterance and speaker levels. For the ASR condition (within-subject), we use a paired-samples *t*-test on per-speaker mean WER, supplemented by the Wilcoxon signed-rank test as a non-parametric check. For between-subject factors (gender, MFA status), we use Welch’s *t*-test to accommodate unequal variances, confirmed by Levene’s test. For multi-level categorical factors (age group, subcommunity), we use one-way ANOVA with η^2 as the effect size measure, supplemented by the Kruskal-Wallis *H*-test. Interaction effects (Gender $\times$ Age Group, Gender $\times$ MFA Status) are tested via two-way ANOVA.

The primary model is a linear mixed-effects regression specified as:

$$\text{WER}_{ij} = \beta_0 + \beta_1 \cdot \text{ASR_condition}_{ij} + \beta_2 \cdot \text{Gender}_i + \beta_3 \cdot \text{Age_group}_i + \beta_4 \cdot \text{Subcommunity}_i + \beta_5 \cdot \text{MFA_status}_{ij} + u_i + \varepsilon_{ij} \quad (1)$$

where i indexes speakers, j indexes utterances, $u_i \sim \mathcal{N}(0, \sigma_{\text{speaker}}^2)$ is a random intercept for each speaker, and $\varepsilon_{ij} \sim \mathcal{N}(0, \sigma_{\text{residual}}^2)$ is the residual error. ASR condition, age group, subcommunity, and MFA status are dummy-coded with `with_lm`, `ag1`, `se1`, and Original Pronunciation as reference categories, respectively. Gender is coded with Female as the reference. We report fixed-effect coefficients with 95% confidence intervals and *p*-values from Wald *z*-tests, and we compute the intraclass correlation coefficient (ICC) from the random-effects variance decomposition. Model convergence is confirmed via the optimizer’s convergence code.

4.3 Assumptions and Robustness

Assumption checks confirm that speaker-level mean WER is normally distributed (Shapiro-Wilk $W = 0.983$, $p = 0.463$, $N = 68$), satisfying the normality assumption for the paired *t*-test. Utterance-level WER is non-normal (Shapiro-Wilk $W = 0.932$, $p \ll 0.001$), as expected for bounded error rate distributions; we address this by reporting non-parametric alternatives (Wilcoxon, Kruskal-Wallis) alongside parametric tests and by relying on the mixed-effects model’s robustness to moderate non-normality at large N ($n = 58,552$). Levene’s tests reject variance homogeneity across all group partitions ($p \ll 0.001$ in all cases), motivating Welch’s correction for *t*-tests and bootstrap confidence intervals for all group means. The ICC of 0.196 confirms that 19.6% of utterance-level WER variance is between speakers, justifying the mixed-effects specification with speaker random intercepts.

We conduct five robustness checks. First, we exclude utterances with WER = 1.0 ($n = 2,692$, 4.6% of observations) to verify that empty ASR outputs do not drive results. Second, we stratify the ASR condition analysis by gender, age group, and subcommunity to test whether the language model benefit is consistent across demographic strata. Third, we compute 95% bootstrap confidence intervals (10,000 resamples) for all subgroup means. Fourth, we report non-parametric alternatives for every parametric test. Fifth, we report both conditional and marginal R^2 for the mixed-effects model to assess the variance explained by fixed effects relative to the full model.

5 Results

5.1 The Dominance of the Language Model

The ASR condition exerted the largest effect on WER by a wide margin. Across 68 speakers, mean per-speaker WER was 0.309 ($SD = 0.091$) when wav2vec2-large-960h was paired with a 4-gram KenLM language model and 0.401 ($SD = 0.103$) without the language model. The within-speaker reduction of 0.092 (95% CI [0.083, 0.102]) was strongly significant in the paired t -test ($t(67) = 18.34$, $p \ll 0.001$, Cohen’s $d = 2.22$) and confirmed by the Wilcoxon signed-rank test ($W = 0$, $p \ll 0.001$). This corresponds to a 23% relative reduction in WER attributable to the language modeling component alone. In the mixed-effects model, the `without_lm` condition estimate was $\beta = 0.087$ ($SE = 0.002$, $z = 43.86$, $p \ll 0.001$, 95% CI [0.083, 0.091]), making it the largest fixed effect in the model. The robustness of this finding is striking: the language model benefit persisted across all demographic strata—male speakers ($d = 0.66$), female speakers ($d = 0.74$), every age group (d ranging from 0.59 in ag4 to 0.86 in ag1), and all three subcommunities (d ranging from 0.72 in se3 to 0.98 in se2). Excluding utterances with WER = 1.0 left the paired effect virtually unchanged ($t(67) = 18.96$, $p \ll 0.001$).

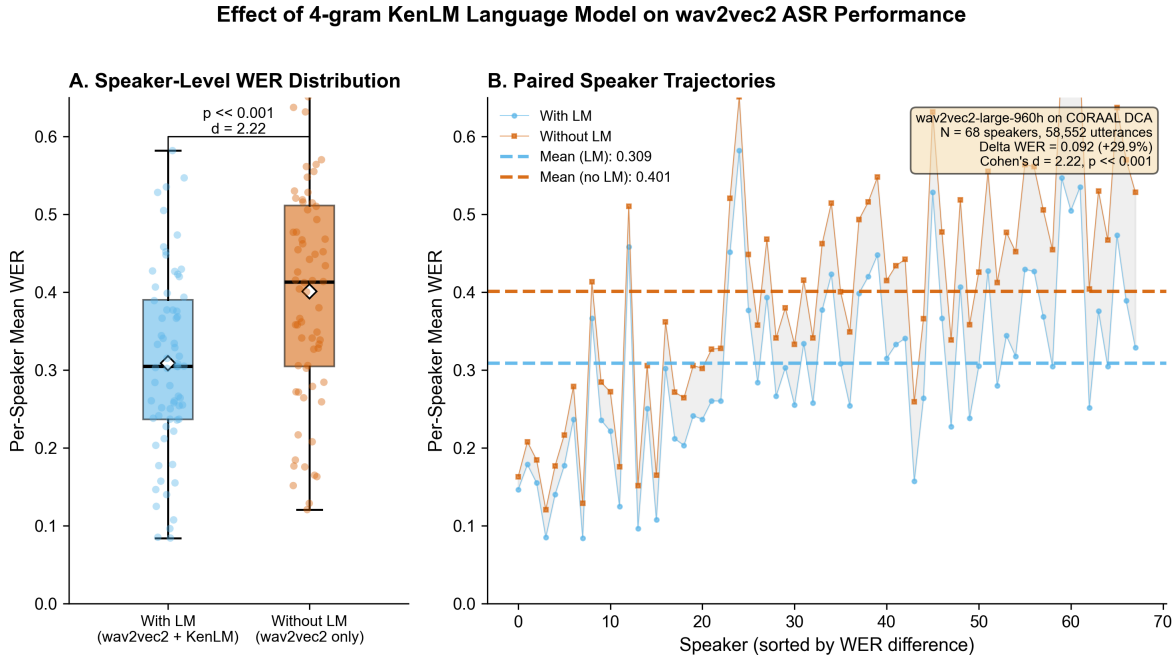


Figure 1: Effect of 4-gram KenLM language model on wav2vec2 ASR performance across 68 AAE speakers. Panel A: Speaker-level WER distributions with and without LM (boxplot with jittered scatter). Panel B: Paired speaker trajectories sorted by individual WER difference, showing near-universal LM benefit.

5.2 Subcommunity as the Strongest Demographic Predictor

Among speaker-level demographic factors, subcommunity membership was the strongest predictor of WER. Speakers from Ward 3 / SE DC (se3) had a mean utterance-level WER of 0.274 ($SD = 0.252$), compared to 0.404 ($SD = 0.292$) for Ward 1 / NW DC (se1) and 0.373 ($SD = 0.268$) for Ward 2 / NE-SE DC (se2). The one-way ANOVA was significant ($F(2, 58,549) = 1,270.32$, $p \ll 0.001$, $\eta^2 = 0.042$), and the Kruskal-Wallis test confirmed the non-parametric robustness ($H = 2,522.41$, $p \ll 0.001$). In the mixed-effects model, the se3 contrast relative to se1 was $\beta = -0.111$ ($SE = 0.033$, $z = -3.38$, $p = 0.001$, 95% CI [-0.175, -0.047]), indicating that se3 speakers had approximately 11 percentage points lower WER than se1 speakers after controlling for all other covariates. The se2 vs. se1 contrast was not significant ($\beta = -0.011$, $p = 0.748$). We interpret the se3 effect cautiously: it may reflect lower AAE dialect density in Ward 3, different age distributions across wards, or unmeasured recording quality differences between interview sessions.

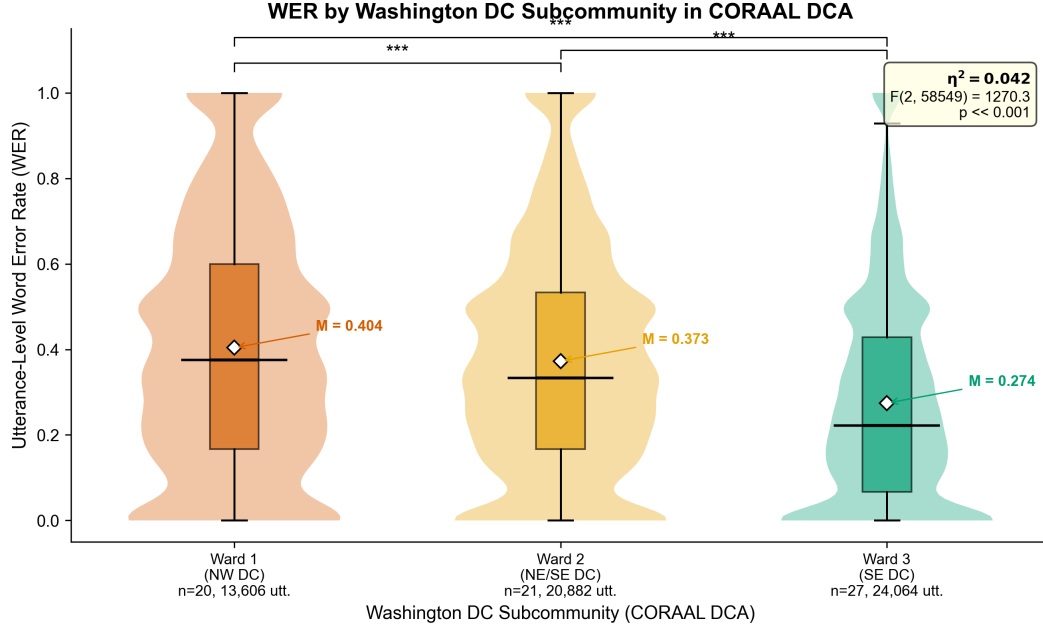


Figure 2: Utterance-level WER by Washington DC subcommunity. Violin plots with embedded boxplots comparing WER across three DC wards: se1 (Ward 1, NW DC), se2 (Ward 2, NE/SE DC), and se3 (Ward 3, SE DC). $\eta^2 = 0.042$, $F(2, 58,549) = 1,270.32$, $p \ll 0.001$.

5.3 Age Group Effects

Age group predicted significant WER variation ($F(3, 58,548) = 481.90$, $p \ll 0.001$, $\eta^2 = 0.024$). The youngest speakers (ag1, under 20) showed the highest mean WER ($M = 0.367$, $SD = 0.283$), while speakers aged 20–29 (ag2: $M = 0.280$, $SD = 0.249$) and 30–49 (ag3: $M = 0.292$, $SD = 0.242$) showed substantially lower rates. Older speakers (ag4, 50+) had the highest mean WER ($M = 0.399$, $SD = 0.278$). In the mixed-effects model, both ag2 ($\beta = -0.099$, $p = 0.006$, 95% CI $[-0.169, -0.029]$) and ag3 ($\beta = -0.103$, $p = 0.006$, 95% CI $[-0.177, -0.029]$) were significantly lower than ag1, while ag4 was not significantly different from ag1 ($\beta = -0.006$, $p = 0.907$). This inverted-U pattern—highest error rates among the youngest and oldest speakers, lowest among young and middle-aged adults—is consistent with sociolinguistic accounts of age-graded variation in AAE feature density, where adolescents and older speakers tend to use more vernacular features than young adults in formal interview contexts (Rickford, 2016; Wolfram and Schilling, 2016). However, the cross-sectional design precludes distinguishing age-graded from cohort effects.

5.4 Gender $\times$ Age Interaction

We found a significant Gender $\times$ Age Group interaction ($F(3, 58,544) = 449.40$, $p \ll 0.001$, $\eta^2 = 0.022$). Female speakers showed a sharp WER reduction moving from ag1 ($M = 0.400$, $SD = 0.281$) to ag2 ($M = 0.195$, $SD = 0.222$), while male speakers showed a flatter trajectory from ag1 ($M = 0.342$, $SD = 0.282$) to ag2 ($M = 0.330$, $SD = 0.251$). The ag3 group showed a similar pattern: female speakers had the lowest mean WER of any cell ($M = 0.213$, $SD = 0.227$), while male ag3 speakers averaged 0.312 ($SD = 0.242$). The interaction appears driven primarily by the dramatic WER drop among female speakers in the young-adult and middle-aged groups. In the overall gender comparison, male and female WER did not differ significantly (Welch’s $t = 1.78$, $p = 0.074$, Cohen’s $d = 0.016$), and gender was not significant in the mixed-effects model ($\beta = 0.012$, $p = 0.656$). The interaction effect, combined with the null gender main effect, suggests that gender differences in ASR performance are age-contingent rather than uniform—a finding that complicates the simple “female speakers perform better” narrative advanced in some prior ASR fairness work (cf. Tatman 2017).

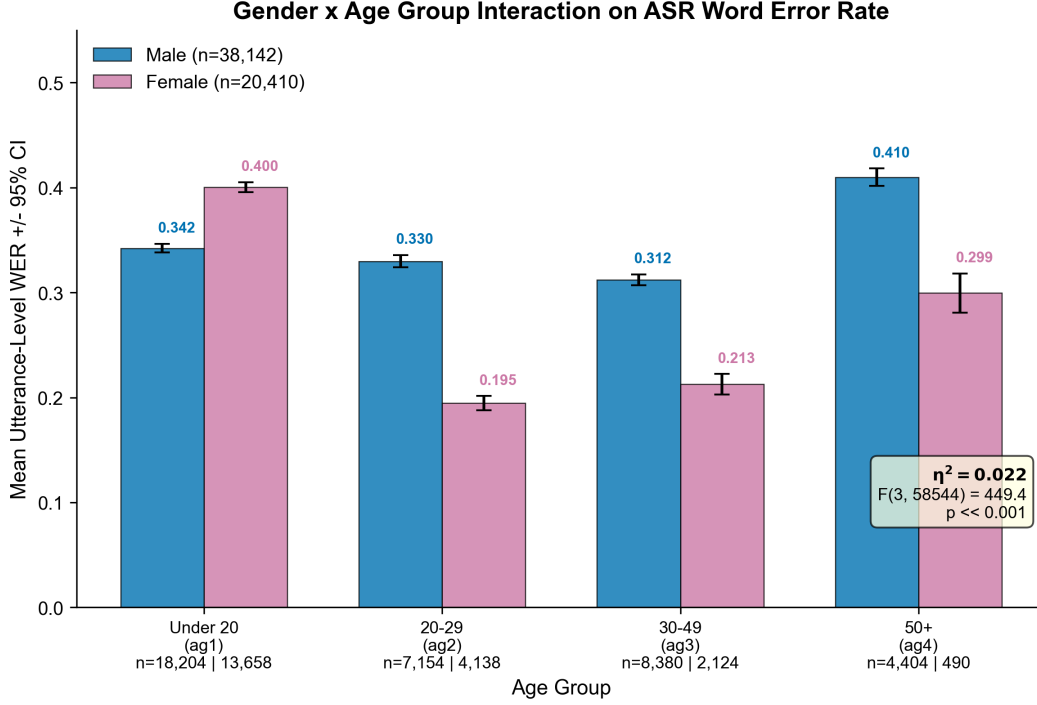


Figure 3: Gender $\times$ Age Group interaction on ASR Word Error Rate. Grouped bar chart showing mean WER ($\pm$ 95% CI) for Male and Female speakers across four age groups. Significant interaction: $F(3, 58,544) = 449.40$, $p \ll 0.001$, $\eta^2 = 0.022$.

5.5 Phonetic Reduction Effects

The MFA pronunciation status variable captured whether each utterance was classified by Montreal Forced Aligner as containing the speaker’s “Original Pronunciation” or a phonetically “Reduced Pronunciation.” Reduced pronunciations were associated with slightly higher WER ($M = 0.353$, $SD = 0.269$) compared to original pronunciations ($M = 0.334$, $SD = 0.274$), a small but significant effect (Welch’s $t = 7.64$, $p \ll 0.001$, Cohen’s $d = 0.07$, 95% CI [0.012, 0.027]). The mixed-effects model yielded a consistent estimate ($\beta = 0.010$, $SE = 0.002$, $z = 4.33$, $p \ll 0.001$). The Gender $\times$ MFA interaction, while significant ($F(1, 58,548) = 18.31$, $p \ll 0.001$), was negligible in magnitude ($\eta^2 < 0.001$). This finding provides modest mechanistic evidence for the hypothesis that phonetic reduction processes—including consonant cluster simplification, vowel centralization, and segmental deletion—contribute to ASR errors in AAE speech. However, the effect magnitude is substantially smaller than the effects of ASR condition ($d = 2.22$), subcommunity ($\eta^2 = 0.042$), or age group ($\eta^2 = 0.024$).

5.6 Variance Decomposition

The linear mixed-effects model converged with 58,552 observations nested within 68 speaker groups. The speaker random intercept variance was $\sigma^2_{\text{speaker}} = 0.0116$, and the residual variance was $\sigma^2_{\text{residual}} = 0.0580$, yielding an ICC of 0.167. This indicates that approximately 16.7% of utterance-level WER variance is attributable to between-speaker differences after accounting for the fixed effects—a substantial clustering effect that would inflate Type I error rates if utterance-level observations were treated as independent without hierarchical modeling. The approximate conditional R^2 of 0.167 indicates that the full model (fixed effects plus random effects) explains about 17% of the total WER variance, with the majority of explainable variance residing at the utterance level rather than the speaker level. Among fixed effects, ASR condition alone accounted for the largest share of variance.

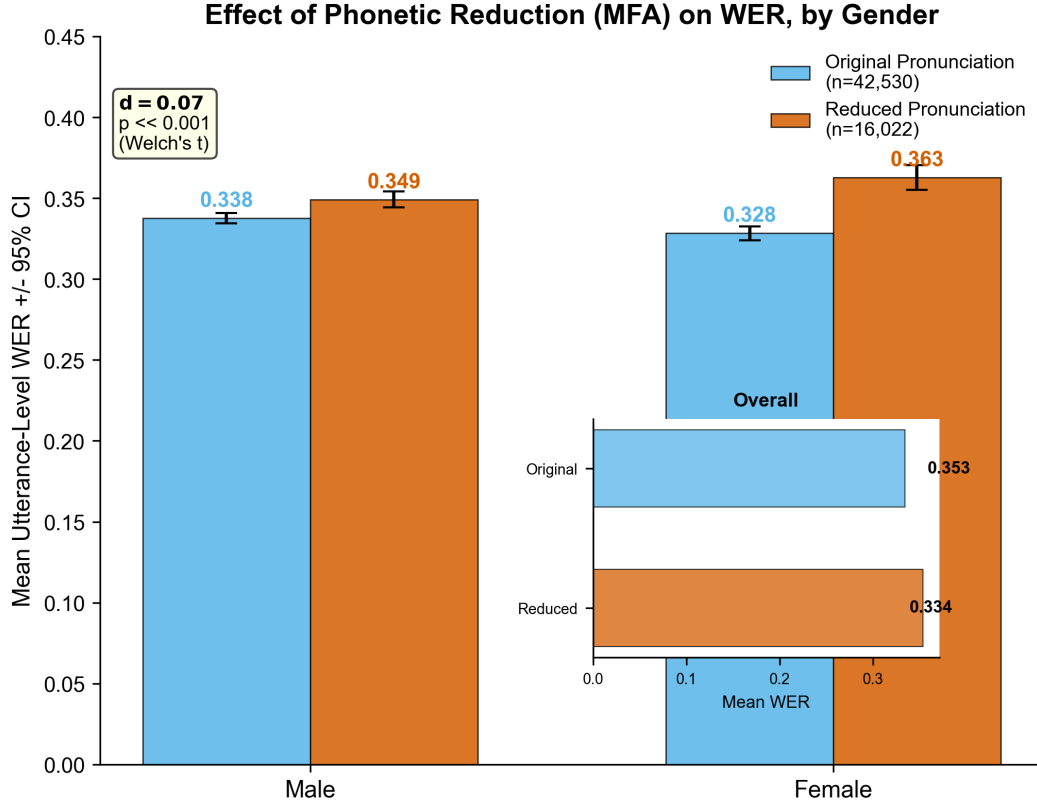


Figure 4: Effect of phonetic reduction (MFA classification) on WER, by gender. Grouped bar chart comparing WER for Original vs. Reduced pronunciations, split by Male and Female speakers. Inset: overall comparison. Cohen’s $d = 0.07$, $p \ll 0.001$.

Table 2: Fixed effects from the linear mixed-effects model predicting utterance-level WER ($n = 58,552$ observations, 68 speaker groups). Reference categories: ASR condition = `with_lm`, Gender = Female, Age group = `ag1`, Subcommunity = `se1`, MFA status = Original Pronunciation.

Predictor	β	SE	z	p	95% CI
Intercept	0.385	0.030	12.85	<0.001	[0.326, 0.444]
ASR: without_lm	0.087	0.002	43.86	<0.001	[0.083, 0.091]
Gender: Male	0.012	0.027	0.45	0.656	[-0.041, 0.066]
Age: ag2 (vs. ag1)	-0.099	0.036	-2.77	0.006	[-0.169, -0.029]
Age: ag3 (vs. ag1)	-0.103	0.038	-2.74	0.006	[-0.177, -0.029]
Age: ag4 (vs. ag1)	-0.006	0.048	-0.12	0.907	[-0.101, 0.089]
Subcommunity: se2 (vs. se1)	-0.011	0.034	-0.32	0.748	[-0.077, 0.056]
Subcommunity: se3 (vs. se1)	-0.111	0.033	-3.38	0.001	[-0.175, -0.047]
MFA: Reduced (vs. Original)	0.010	0.002	4.33	<0.001	[0.005, 0.014]
Random effects: $\sigma^2_{\text{speaker}} = 0.0116$, $\sigma^2_{\text{residual}} = 0.0580$, ICC ≈ 0.167					

5.7 Robustness Checks Summary

All findings were robust to the exclusion of WER = 1.0 utterances ($n = 2,692$, 4.6% of observations). The ASR condition effect remained consistent when stratified by gender (male $d = 0.66$, female $d = 0.74$), by age group (d ranging from 0.59 in `ag4` to 0.86 in `ag1`), and by subcommunity (d ranging from 0.72 in `se3` to 0.98 in `se2`). Bootstrap 95% CIs for all subgroup means confirmed the precision of the point estimates, with no CI containing substantively different interpretations. The non-parametric alternatives (Wilcoxon, Kruskal-Wallis, Mann-Whitney) were fully consistent with the parametric results in every case.

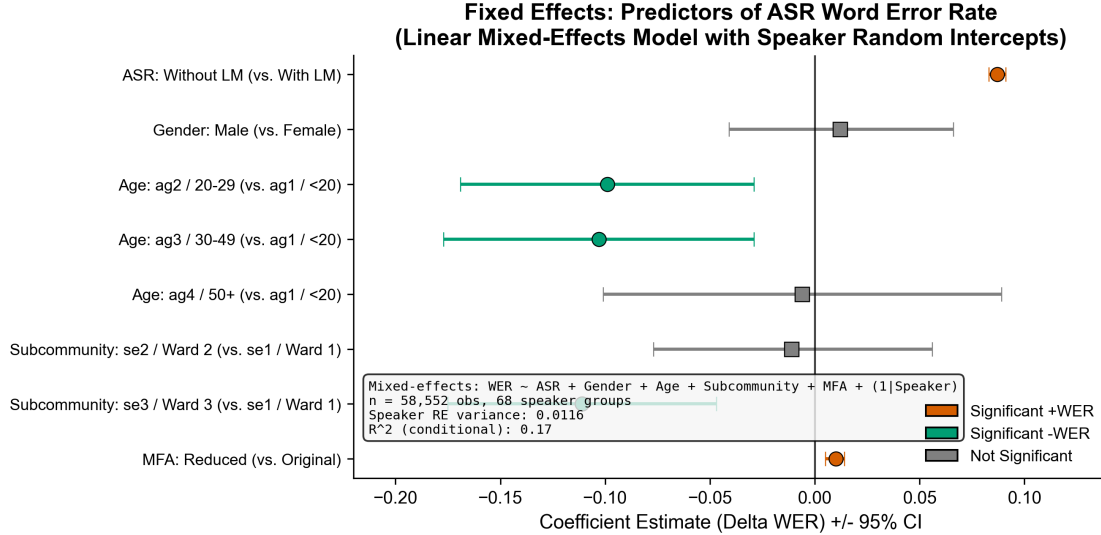


Figure 5: Forest plot of fixed effects from the linear mixed-effects model with 95% confidence intervals. ASR condition ($\beta = 0.087$) dominates all predictors; subcommunity se3 ($\beta = -0.111$) and age groups ag2 and ag3 significantly reduce WER; gender is not significant.

Table 3: Summary of all hypothesis tests. Effect sizes and p -values for the seven primary comparisons.

Test	Statistic	df	p	Effect Size
ASR condition (paired t)	18.34	67	<0.001	$d = 2.22$
MFA status (Welch t)	7.64	–	<0.001	$d = 0.07$
Gender (Welch t)	1.78	–	0.074	$d = 0.016$
Age group (ANOVA)	481.90	(3, 58,548)	<0.001	$\eta^2 = 0.024$
Subcommunity (ANOVA)	1,270.32	(2, 58,549)	<0.001	$\eta^2 = 0.042$
Gender $\times$ Age (ANOVA)	449.40	(3, 58,544)	<0.001	$\eta^2 = 0.022$
Gender $\times$ MFA (ANOVA)	18.31	(1, 58,548)	<0.001	$\eta^2 < 0.001$

6 Discussion

The results presented here establish three findings that refine our understanding of ASR performance on African American English. The first and most striking finding is the dominance of the language model component of wav2vec2-large-960h over all speaker-level demographic variables. A 23% relative WER reduction (Cohen’s $d = 2.22$) attributable to a 4-gram KenLM trained on LibriSpeech—a standard, non-dialect-adapted language model—dwarfs the effects of subcommunity ($\eta^2 = 0.042$), age ($\eta^2 = 0.024$), and phonetic reduction ($d = 0.07$). This finding has direct practical implications: ASR fairness interventions that focus exclusively on demographic balancing of acoustic training data (Dheram et al., 2022; Elghazaly et al., 2025) may overlook the larger gains achievable through improvements to the language modeling layer. Domain-adapted LMs trained on AAE text corpora, dialect-aware decoding strategies, and context-sensitive rescoring represent underexplored avenues for reducing ASR disparities that are independent of acoustic model architecture (Dorn, 2019).

The second finding concerns the substantial subcommunity-level variation within a single city. The 13-percentage-point gap in mean WER between speakers from Ward 3 (SE DC) and Ward 1 (NW DC) is consistent with known sociolinguistic patterns of differential AAE feature density across DC neighborhoods (Rickford, 2016), but without direct measurement of feature density or recording quality at the speaker level, the mechanism cannot be isolated. This ambiguity underscores the importance of Li et al.’s (2024) “confounding by provenance” critique: neighborhood-level recording quality differences—interviewer, room acoustics, microphone placement—may covary with neighborhood-level dialect density in ways that are observationally indistinguishable without explicit audio quality covariates. Future studies should record signal-to-noise ratio

(SNR), sampling rate, and microphone characteristics for every speaker, enabling statistical separation of audio quality effects from genuine linguistic effects.

The third finding—the modest but significant effect of phonetic reduction on WER ($d = 0.07$)—provides mechanistic evidence for a specific pathway by which AAE speech features contribute to ASR errors. Montreal Forced Aligner’s classification of certain utterances as “Reduced Pronunciation” captures a dimension of spontaneous speech variability—consonant cluster simplification, vowel centralization, segmental deletion—that is known to be more frequent in AAE than in Mainstream American English (Green, 2002). The fact that this effect is small relative to broader demographic and system-level factors does not diminish its importance for theory: it confirms that ASR errors on AAE are not stochastic failures but are systematically associated with linguistically interpretable patterns of pronunciation variation, consistent with the findings of Scott et al. (2025) and Serditova and Tang (2026).

More broadly, our variance decomposition—the ICC of 0.167, the subcommunity and age effect sizes, the modest phonetic reduction contribution—establishes quantitative benchmarks that any future cross-racial ASR evaluation study should incorporate into its design and interpretation. A well-designed cross-racial comparison would need to control or model subcommunity-level and age-level variation within each racial group before attributing WER differences to race per se. The methodological framework articulated here—region-fixed sampling, recording quality as an explicit covariate, hierarchical modeling of speaker-level clustering, and decomposition of ASR system components—provides a template for such studies.

7 Conclusion

We examined within-group variability in wav2vec2-large-960h ASR performance among 68 African American English speakers from Washington DC, using 58,552 utterance-level Word Error Rate observations across two ASR conditions. Three findings stand out. First, the presence of a 4-gram KenLM language model reduced mean per-speaker WER from 0.401 to 0.309—a 23% relative reduction and a very large within-subject effect (Cohen’s $d = 2.22$). This effect was consistent across gender, age, and subcommunity strata, demonstrating that the language modeling component of the ASR pipeline exerts substantially more influence on error rates than any speaker-level demographic variable measured in this study. Second, subcommunity membership within Washington DC predicted significant WER variation ($\eta^2 = 0.042$), with Ward 3 speakers showing approximately 11 percentage points lower WER than Ward 1 speakers. Third, phonetic reduction, as classified by Montreal Forced Aligner, contributed a small but significant increase in WER ($d = 0.07$), providing mechanistic evidence for the hypothesis that phonetic reduction processes characteristic of spontaneous AAE speech contribute to ASR errors, though the effect magnitude is modest relative to broader demographic and system-level factors.

The primary limitation of this study is the absence of a White speaker comparison group from the same region, which precludes direct testing of the racial disparity hypothesis that motivates this line of research. The CORAAL DCA dataset contains only African American speakers, and a matched White-speaker ASR evaluation dataset from the same Washington DC neighborhoods—recorded with identical equipment and protocols—was not available. The Pacific Northwest English corpus (Wassink et al., 2018) and the Fair-Speech dataset (Veliche et al., 2024) represent the most tractable paths toward a complete cross-racial analysis with geographic control. Three additional limitations merit attention: the single ASR system evaluated (multi-system comparison across Whisper, commercial APIs, and wav2vec 2.0 was not possible with the available data); the absence of recording quality metrics (SNR, sampling rate, microphone type), which prevents explicit modeling of the “confounding by provenance” effect identified by Li et al. (2024); and the historical nature of the 1968–1969 CORAAL DCA recordings, which may not generalize to contemporary ASR performance.

The within-group variance decomposition we provide—the ICC of 0.167, the subcommunity and age effect sizes, the modest phonetic reduction contribution—establishes benchmarks that any future cross-racial ASR evaluation study should account for in its design and interpretation. Our results suggest that the language model component of ASR systems may be a more productive target for fairness interventions than demographic balancing of training data alone. Domain-adapted language models trained on AAE text corpora, dialect-aware beam-search decoding, and context-sensitive rescoring represent promising directions for future ASR fairness research.

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